

# DAVID DING

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## **OBJECTIVE**

As a Golang Full Stack developer with over 8 years of experience in Golang backend development, my primary goal is to meet client and customer needs through my technical expertise, domain knowledge, and collaborative attitude.

## **SUMMARY**

- Proficient in designing and building large-scale distributed systems and cloud-native microservices, supporting high-throughput transaction processing and internal platform services.
- Experienced leveraging AI-assisted development tools (e.g., Cursor, Claude Code) to accelerate development, refactoring, test, debugging, improving engineering productivity and code quality.
- Experienced leveraging Go frameworks and ecosystem tools (Gin, Echo, Go Kit, GORM, Testify) to production-level RESTful APIs and microservices aligned with enterprise engineering standards.
- Strong ownership across the full SDLC, including system design, implementation, testing, CI/CD deployment, and operational support, with familiarity in secure secrets management and encryption practices (e.g., Vault/KMS concepts) in cloud-native environments.
- Proven experience architecting event-driven microservices and high-throughput pipelines using Kafka and queue-based systems, with familiarity in rule-based detection concepts (e.g., YARA/YARA-X) and real-time data processing.
- Hands-on experience leading frontend-backend collaboration, ensuring contract alignment and performance across React- and AngularJS-based web applications.
- Skilled in implementing DevOps practices and CI/CD pipelines using GitHub Actions, Jenkins, uDeploy, and SonarQube to enable automated testing, build, and deployment processes.
- Strong expertise in distributed systems debugging and performance optimization, goroutine analysis, memory profiling, GC tuning, and latency reduction for high-throughput services.
- Experienced designing scalable data storage and access layers using relational databases (PostgreSQL, MySQL) and NoSQL/cache systems (Redis, MongoDB) for transactional workloads.
- Demonstrated ability to deliver RESTful backend services from proof of concept through production deployment, partnering closely with product, UX, and platform engineering teams.
- Experience with platform and infrastructure teams to improve deployment workflows, automate service delivery pipelines, simplify Kubernetes-based service deployments for engineering teams.

## **VISA STATUS: U.S. Citizen**

## **TECHNICAL SKILLS**

Languages: Golang 1.10 – 1.24, Java 1.0-5,7, 8-24, JavaScript (ES6+); TypeScript, Python (basic scripting)

Databases: Oracle, MySQL, PostgreSQL, SQL server, MongoDB, Redis, Cassandra.

Frameworks: Spring MVC, Spring Boot 2-3, Angular 12-18, React 16-18, Node.js 16, 18, 20

DevOps: Docker, GitHub Actions, Jenkins, uDeploy, SonarQube

Cloud: Kubernetes (EKS), Prometheus, OpenTelemetry, CloudWatch, Splunk

## **EDUCATION BACKGROUND**

### **Ph.D. in Chemical Engineering**

Lamar University, Beaumont, Texas. GPA 3.80 /4.00

**Aug 2011 -- May 2015**

### **B.E. in Chemical Engineering**

East China University of Science and Technology, Shanghai, China

**Aug 2007 -- May 2011**

## **CERTIFICATES:**

AWS Certified Developer – Associate, Amazon, April 2025

AWS Certified DevOps Engineer – Professional, May 2025

AWS Certified AI Practitioner – Foundational, May 2025



## **PROFESSIONAL EXPERIENCES**

**American Express, NYC, NY**

**Feb 2022 – Present**

**Senior Golang Developer**

**Client Fulfillment Onboarding (CFO)/MySetup:** Backend services supporting client onboarding, identity validation, and policy-driven eligibility checks. Responsible for maintaining and modernizing multiple Golang services related to client profile evaluation and workflow execution.

**Mobile SRE team:** Worked closely with customer service/mobile engineering teams to analyze production issues, trace services dependencies, provide initial triage for distributed system failures.

One Identification Task Force: Maintained and enhanced **Golang-based auth modules** used across Amex web and mobile applications, improving throughput, consistency, and cross-system data integrity.

### **Responsibilities**

- Maintained and modernized Golang microservices powering client onboarding and identity verification workflows, incorporating rule-based validation and pattern-matching logic for real-time data processing.
- Designed and evolved RESTful APIs and backend data models to align with frontend requirements, ensuring smooth integration with ReactJS/AngularJS-based UI flows.
- Deployed containerized Golang services to AWS EKS (Kubernetes) and collaborated with platform teams to improve deployment stability and service scalability.
- Implemented structured logging, distributed tracing, and metric dashboards (Splunk, Prometheus, CloudWatch) to provide deep visibility into service behavior.
- Supported troubleshooting of high-throughput distributed pipelines by analyzing logs, traces, and metrics, improving system reliability, backpressure handling, and real-time data processing efficiency.
- Collaborated with platform teams to streamline containerized service deployments and improve internal deployment workflows for Golang microservices running on Kubernetes.

**Bank of America, Pennington, NJ**

**Oct 2019 -- Feb 2022**

**Golang Developer**

**BOFA Trading Account Application:** Worked on backend services for the Trading Account Payment Platform, which supported payments, refunds, split settlements, and multi-gateway routing (PayPal, Visa, Stripe). The system included a unified payment engine, real-time transaction tracking, automated reconciliation workflows.

### **Responsibilities**

- Developed and maintained distributed Golang microservices for payment authorization, refund workflows, settlement routing, and transaction status propagation across distributed systems.
- Designed and optimized idempotent REST APIs and backend processing logic, with familiarity in rule-based detection concepts (e.g., YARA/YARA-X) for handling transaction validation and anomaly detection scenarios.
- Built event-driven pipelines using Kafka/queue-based systems to support multi-step payment workflows, incorporating fraud detection, rule-based validation, and real-time data processing.
- Leveraged both SQL (PostgreSQL/MySQL) and NoSQL technologies to support high-throughput transaction storage, real-time status queries, and reconciliation workloads.
- Contributed to monitoring and alerting using CloudWatch, Splunk, and custom metrics to track transaction flow health and detect workflow bottlenecks.

**Brown Brothers Harriman, Jersey City, NJ**

**Nov 2018 -- July 2019**

**Java Developer**

**Cyber Security Monitoring:** Built a vulnerability detection and processing system that ingested reports from external vendors (Qualys), applied rule-based parsing and pattern matching to identify security issues, and integrated with centralized ticketing systems (Jira) for automated alerting and remediation workflows.

- Developed backend services to ingest and process security data via scheduled jobs (Quartz/CRON), enabling rule-based analysis and automated vulnerability detection workflows.
- Analyzed security data (Splunk) to identify patterns anomalies, supporting detection workflows.
- Implemented notification features ensuring timely delivery of vulnerability detection results to relevant teams.

**Chemical/Pharmaceutical Related Job**

**May 2015 -- June 2018**

**Quality Control Analyst**

Worked in Laboratory for pharmaceutical/supplemental product tests.